Home Lab Project Week 10, Day 1 new

Create targeted firewall rules that allow only the required protocols between designated lab hosts, and restrict inbound access to the vulnerable Ubuntu VM.

Task phrase	What it means in your lab
“Allow only the required protocols between designated lab hosts”	Kali is the only host allowed to reach Ubuntu, and only on SSH (22), HTTP (80), and ping
“Restrict inbound access to the vulnerable Ubuntu VM”	**Every other** path into Ubuntu is blocked: other ports, other hosts, and the internet (WAN)

My earlier rule 3 only blocked *Kali's* extra traffic. To meet the second part, the block rule should protect Ubuntu from **any** source.

Revised LAN rules

Order	Action	Protocol	Source	Destination	Port	Log	Description
1	Pass	TCP	Kali_Host	Ubuntu_Host	Kali_to_Ubuntu_Ports	Off	Allow Kali SSH/HTTP to Ubuntu
2	Pass	ICMP (Echo request)	Kali_Host	Ubuntu_Host	—	Off	Allow Kali ping to Ubuntu
3	Block	any	**any**	Ubuntu_Host	any	**On**	Default deny inbound to Ubuntu

Rule 3 now stops anything arriving on LAN from reaching Ubuntu unless rules 1 or 2 already allowed it. That covers Kali on other ports and any other LAN device you might add later. Logging it gives you evidence that the deny is working.

Check the WAN side

The internet is the other inbound path. You don't need new rules here, just proof that nothing is open:

1. **Firewall → NAT → Port Forward** should have **no** entries pointing to `192.168.20.10`. A port forward is the only way outside traffic would be sent to Ubuntu.
2. **Firewall → Rules → WAN** should have no pass rules to the DMZ. OPNsense blocks unsolicited WAN traffic by default.

Take a screenshot of each page, since "nothing is configured" is your evidence here.

Updated tests from Kali

Test	Command	Expected	Proves
Ping	`ping -c 4 192.168.20.10`	Replies	ICMP allowed
SSH	`nc -zv -w 3 192.168.20.10 22`	Succeeded	SSH allowed
HTTP	`nc -zv -w 3 192.168.20.10 80`	Succeeded or refused	HTTP allowed through firewall
Other port	`nc -zv -w 3 192.168.20.10 3306`	Times out	Default deny works
Port sweep	`nmap -Pn -p 1-1000 192.168.20.10`	Only 22 (and 80) not "filtered"	Nothing else reachable

The nmap sweep is strong evidence for "restrict inbound access," because it shows the whole picture in one screenshot. Filtered ports mean the firewall dropped the probes.

Optional: defense in depth

Because the task calls Ubuntu *vulnerable*, you can earn extra credit by adding a host firewall on Ubuntu as a second layer. If OPNsense is ever misconfigured, Ubuntu still protects itself:

bash
```
sudo ufw default deny incoming
sudo ufw allow from 192.168.10.10 to any port 22 proto tcp
sudo ufw allow from 192.168.10.10 to any port 80 proto tcp
sudo ufw enable
sudo ufw status verbose
```